

Report

5/8/2025

Outline

- Most LLMs struggle to count letters/words
 - Why Do Large Language Models (LLMs) Struggle to Count Letters?
- Most LLMs Lack Character-level understanding
 - Large Language Models Lack Understanding of Character Composition of Words
- Placing the Counting Problem in the Chomsky Hierarchy
 - NEURAL NETWORKS AND THE CHOMSKY HIERARCHY
 - Autoregressive + Chain of Thought $\approx$ Recurrent: Recurrence's Role in Language Models' Computability and a Revisit of Recurrent Transformer
- Theoretical Guarantee: Transformer + CoT Can Solve Counting Tasks
 - Chain of Thought Empowers Transformers to Solve Inherently Serial Problems
 - Counting Ability of Large Language Models and Impact of Tokenization
 - LLM The Genius Paradox: A Linguistic and Math Expert's Struggle with Simple Word-based Counting Problems

Motivation

- Most LLMs cannot precisely count the letters in a word/string
 - How many r's are there in “strawberry” □ (Once a challenge for many LLMs, but most can now handle it correctly.)
- This challenge becomes even more pronounced when LLMs are asked to count the number of words in a given passage

Most LLMs struggle to count letters/words

Why Do Large Language Models (LLMs) Struggle to Count Letters?

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Background and Motivation

- LLMs excel in complex tasks but fail at simple ones like counting letters of a word
- Example: Counting "r"s in "strawberry" is often inaccurate
- This paper explores why LLMs struggle with counting letters

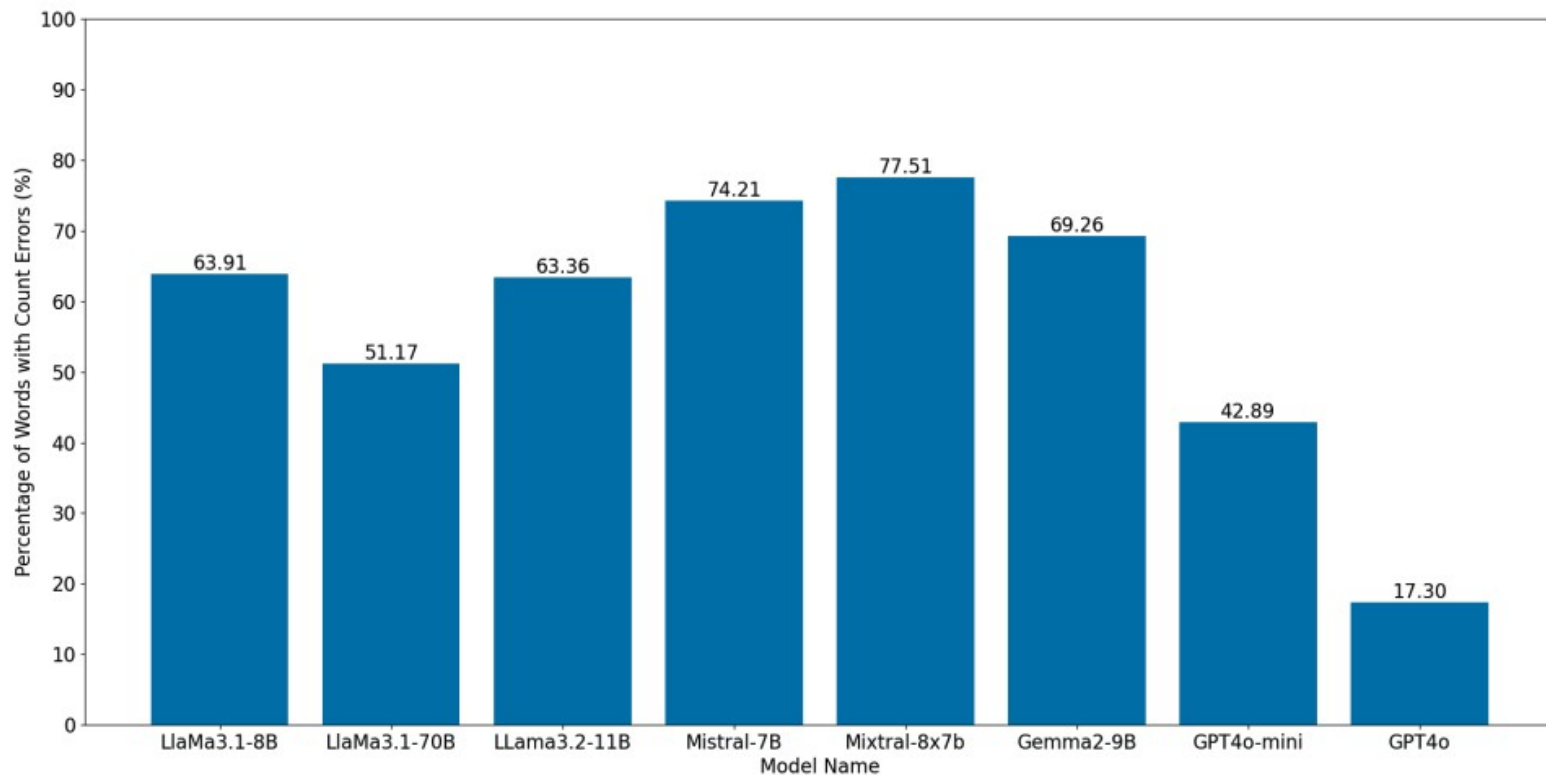
Experimental Settings

- LLMs
 - Meta: LLaMA3.1-8B, LLaMA3.1-70B, LLaMA3.2-11B
 - Mistral: Mistral-7B, Mixtral-8x7B
 - Google: Gemma-2-9B
 - OpenAI: GPT-4o-mini, GPT-4o
- Dataset: 10,000 random words from a 333,000-word list

Task

- Task Description: LLMs must count the frequency of each letter in a given word and return the result as a JSON object
- Example:
 - Input: strawberry
 - Correct Output: {"s": 1, "t": 1, "r": 3, "a": 1, "w": 1, "b": 1, "e": 1, "y": 1}

Error Rates Across Models



Features possibly Influencing LLM Counting Ability

- Frequency Features
 - Word frequency
 - Token frequency
 - Letter frequency
- Count Difficulty Features
 - Number of letters
 - Number of tokens
 - Letter multiplicity (e.g., repeated letters)
- Tokenization

Correlation between Frequency Features and Errors

	LlaMa3.1-8B	LlaMa3.1-70B	LLaMa3.2-11B	Mistral-7B	Mixtral-8x7B	Gemma2-9B	GPT4o-mini	GPT4o
Word Frequency	-0.04	-0.04	-0.03	0.02	-0.02	0.01	-0.04	-0.05
Average Frequency of Tokens	-0.12	-0.09	-0.12	-0.15	-0.11	-0.09	-0.08	-0.09
Number of Letters	0.55	0.59	0.56	0.42	0.40	0.42	0.58	0.44
Difference between Number of Letters and Distinct Letters	0.72	0.69	0.61	0.45	0.48	0.47	0.66	0.50
Number of Tokens	0.27	0.30	0.28	0.14	0.16	0.16	0.29	0.19
Difference between Letters and Distinct Letters that appear on several tokens	0.61	0.62	0.53	0.37	0.41	0.38	0.58	0.44

Tokenization's Limited Role

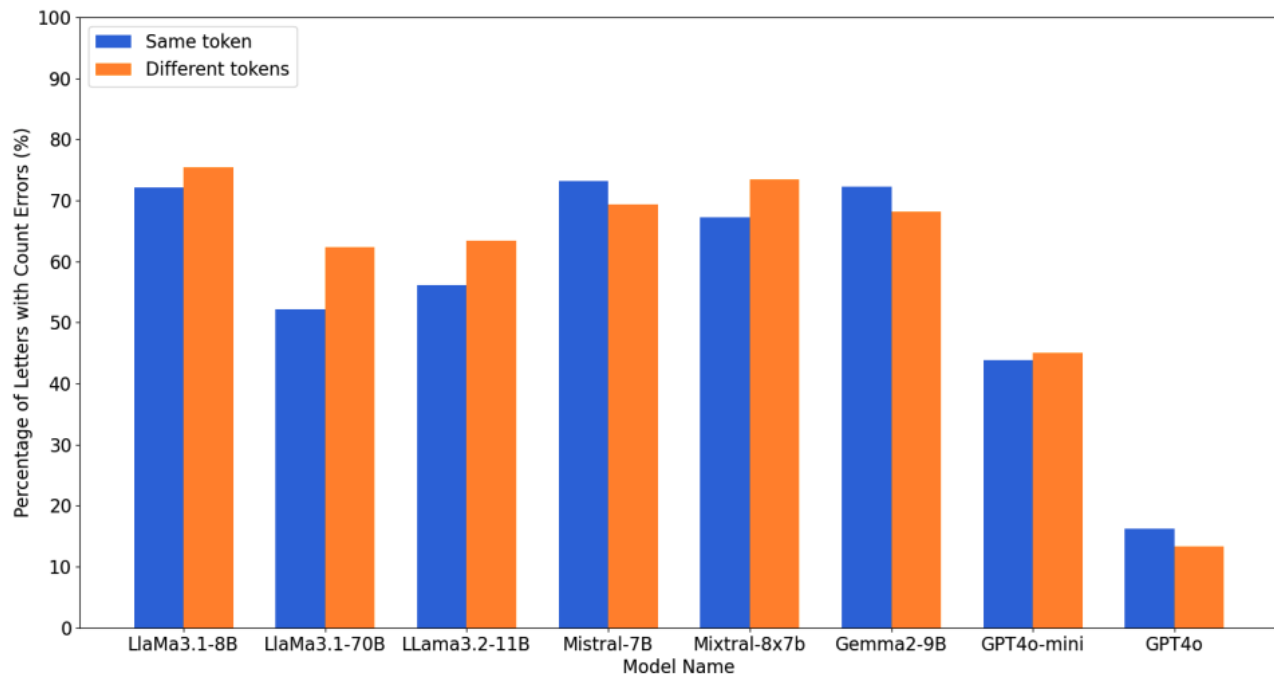


Fig. 10. Percentage of errors when counting a letter that appears twice on a word when a) the two occurrences are in the same token and b) in different tokens

Conclusion

- Errors increase with counting difficulty, especially repeated letters.
- Word/token frequency doesn't affect errors much

Q & A